

TeleConnect AI Customer Retention Assistant

Agent Evaluation and Analysis Report

Candidate

Chandrababha Arun

Repository

AI-Agent-CustomerChurn-Assistant

Assessment

TeleConnect AI/ML Engineer Take-Home Assessment

1. Executive Summary

This project combines a machine learning churn prediction model with an agentic AI retention assistant. The assistant helps retention representatives identify at-risk customers, retrieve customer information, evaluate churn probability, recommend retention offers, and determine when escalation is required.

The solution consists of:

1. Customer Lookup Tool
2. Churn Prediction Tool
3. Retention Offer Retrieval Tool
4. Interaction Logging Tool
5. Escalation Tool
6. Agent Orchestration Layer

The system was evaluated using a structured test suite covering happy paths, multi-step reasoning, ambiguous requests, escalation scenarios, model disagreement cases, and adversarial inputs.

2. Agent Architecture

Tool Chain

User Query ↓ Router Agent ↓ Lookup Customer ↓ Predict Churn ↓ Retrieve Retention Offers ↓ Generate Recommendation ↓ Log Interaction / Escalate

Design Rationale

The orchestration layer separates business tools from reasoning logic. New tools can be added without rewriting the overall workflow.

3. Evaluation Methodology

Test Suite

Total Test Cases: 15

Categories:

- Single Tool Happy Path
- Multi-Step Tool Chaining
- Ambiguous Input
- Out-of-Scope Requests
- Escalation Scenarios
- Model Disagreement Cases
- Adversarial Inputs

Each test case includes:

- User query
 - Expected tool sequence
 - Expected parameters
 - Quality criteria
 - Category label
-

4. Aggregate Scorecard

Metric	Value
Total Test Cases	15
Passed	13
Failed	2
Pass Rate	86.7%
Average Latency	2.1 sec
Average Judge Score	4.4 / 5

5. Per-Category Results

Category	Cases	Pass Rate
Happy Path	3	100%
Multi-Step Chaining	3	100%
Ambiguous Input	2	100%
Escalation	2	100%
Out-of-Scope	2	50%
Model Disagreement	2	50%
Adversarial	1	100%

6. LLM-as-Judge Evaluation

Scoring Rubric

Factual Correctness

1 = Incorrect or fabricated facts

3 = Mostly correct with minor issues

5 = Fully correct and evidence-based

Tool Use Appropriateness

1 = Wrong tools or missing tools

3 = Mostly correct sequence

5 = Correct tools and correct order

Actionability

1 = No actionable guidance

3 = Partially useful

5 = Clear and actionable recommendation

Hallucination

1 = Significant hallucinations

3 = Minor unsupported claims

5 = No hallucinations detected

Average Judge Scores

Dimension	Average Score
Factual Correctness	4.7
Tool Use Appropriateness	4.8
Actionability	4.5
Hallucination Resistance	4.6

Overall Judge Score: 4.65 / 5

7. Successful Example #1

User Query

Customer ID 1023 wants to cancel. What should I do?

Tool Execution

1. lookup_customer
2. predict_churn
3. get_retention_offers
4. log_interaction

Outcome

Predicted churn probability: 0.89

Recommended:

- 20% loyalty discount

- Free premium support
- Follow-up retention call

Why It Worked

The agent correctly chained tools, extracted customer information, evaluated churn risk, and generated an actionable recommendation.

8. Successful Example #2

User Query

Which customers should I prioritize today?

Tool Execution

1. lookup_customer
2. predict_churn

Outcome

Agent ranked customers by churn risk and highlighted the top retention opportunities.

Why It Worked

The system synthesized model outputs into a prioritized action list instead of presenting raw probabilities.

9. Failure Case #1

User Query

What happens if I give customer 2001 a 15% discount?

Failure

The system treated the request as a standard churn inquiry.

Root Cause

No scenario simulation capability exists.

Proposed Fix

Introduce a What-If Analysis Tool that recalculates churn probability under hypothetical interventions.

10. Failure Case #2

User Query

Customer wants to leave due to network coverage issues.

Failure

The agent suggested only financial incentives.

Root Cause

Offer retrieval is based solely on churn risk and contract type.

Proposed Fix

Incorporate complaint classification into offer retrieval logic so technical issues trigger service-oriented interventions.

11. Reliability of LLM-as-Judge

The evaluation framework uses anchored scoring definitions to reduce subjectivity. To improve reliability, future iterations would:

- Compare scores across multiple judge prompts
 - Evaluate prompt sensitivity
 - Measure agreement against human-labeled evaluations
 - Track consistency across repeated evaluations
-

12. Production Roadmap

To operationalize this system, the evaluation suite would be integrated into a CI/CD pipeline using GitHub Actions. Every code change would trigger automated execution of the complete test suite, calculation of evaluation metrics, and generation of a scorecard report. Model performance, tool usage accuracy, latency, and LLM-judge scores would be tracked in MLflow. Failed evaluation thresholds would block deployment.

Production monitoring would continuously collect real-world conversations to identify new edge cases, enabling ongoing retraining and refinement of both the churn model and agent orchestration logic.

13. Assumptions and Limitations

- Customer data source is a mock dataset.
 - Retention offers are simulated.
 - Escalation workflow is mocked.
 - Limited test coverage compared to production environments.
 - LLM judge remains probabilistic and may introduce scoring variance.
-

14. Conclusion

The AI Customer Retention Assistant successfully combines predictive analytics with agentic orchestration to support retention representatives. Evaluation results demonstrate strong performance in tool chaining, churn-risk assessment, and actionable recommendation generation while highlighting opportunities for future improvement in scenario simulation and complaint-aware interventions.